

# Linux ACL Basics

1 ACL is superset of regular permissions.

First, create a directory...

```
$ mkdir dir
```

Regular Permissions:

```
$ ls -ld dir
drwx-----
```

ACL Permissions

```
$ getfacl dir
USER: traw rwx
GROUP: traw ---
OTHER: ---
```

Both permissions are the same, let's change that...

3 The "mask" acts as an upper bound, making sure regular permissions are still respected:

Let's remove the group permission, old school way

```
$ chmod g=-r dir
```

Regular Permissions:

```
$ ls -ld dir
drwx-----+
```

ACL Permissions:

```
$ getfacl dir
USER: traw rwx
user: jay R--
GROUP: traw ---
mask: ---
OTHER: ---
```

Now "jay" can't read the directory. Its permission is uppercase - meaning it's no longer effective!

2 ACL allows adding permission for specific users and groups

Regular Permissions:

```
$ ls -ld dir
drwxr-----+
```

ACL Exists

This is now the "mask" aka the union of:

- GROUP,
- Any custom ACL user
- Any custom ACL group

ACL Permissions:

```
$ getfacl dir
USER: traw rwx
user: jay r--
GROUP: traw ---
mask: r--
OTHER: ---
```

The "mask" is the mechanism that keeps "regular permissions" and ACL in sync  
It's updated automatically when running 'chmod' or 'setfacl'



|              |            |   |     |             |
|--------------|------------|---|-----|-------------|
| owner        | user::rw-  | ↔ | rw- | owner class |
| owning group | group::r-- | ↔ | r-- | group class |
| other        | other::--  | ↔ | --- | other class |

```
$ setfacl -m u:jimy:r file
```

|              |               |   |     |             |
|--------------|---------------|---|-----|-------------|
| owner        | user::rw-     | ↔ | rw- | owner class |
| named user   | user:jimy:rw- |   |     |             |
| owning group | group::r--    | ↔ | r-- | group class |
| mask         | mask::rw-     | ↔ | --- | other class |
| other        | other::--     |   |     |             |